Power Architecture Derivations

Relative Institutional Packages and Type-Dependent Participation

Working formal appendix

2026-05-11

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1 Status

This is the clean working surface for the reimplemented model architecture. The prior feasibility-branch derivation has been archived in git tag `redesign-feasibility-branch-2026-05-11`. Do not import theorem statements, notation, or branch labels from that archived architecture without rederiving them here.

The active manuscript `formal_model_v5.Rmd` remains frozen. New formal work should be developed in this file first, checked by scripts, independently verified, and only then transported to the paper.

1.1 Architecture Decision

The main baseline no longer uses fixed transfers that may fit one realized pie but not another. Proposals are relative institutional packages: quota rules, production shares, cut obligations, enforcement provisions, exceptions, and flexibility terms. These packages are feasible in every state.

Screening comes from `H`'s type-dependent participation threshold:

$$U_H(y, \theta) = y + b_H(\theta),$$

where y is the concession offered by weak states and $b_H(\theta)$ is the direct benefit that `H` obtains from being inside the agreement. `H` accepts if the package payoff exceeds its continuation value.

2 Q&A

What is the substantive puzzle?
Why would a powerful hegemon choose unanimity or consensus when that rule appears to constrain its formal power?

What mechanism should the model isolate?
Unanimity can make an informed hegemon pivotal. Weak states must offer a package that satisfies `H`'s participation constraint, but they do not observe how favorable the package must be for `H` to prefer agreement to its outside option.

What is the key modeling reset?
The proposal is a relative institutional package, not an absolute transfer. It is always feasible. The private-information problem is about `H`'s type-specific acceptance threshold.

What is the OPEC interpretation?
Members bargain over quotas, cuts, shares, monitoring, exceptions, and flexibility. They can observe the proposed package, but they do not know Saudi Arabia's true participation threshold because Saudi Arabia privately knows its spare capacity, opportunity cost, and outside option.

3 Follow-up Q&A for This Proof Pass

Does the clean baseline use weak-state recognition in both rounds? Yes. For this proof pass, $\pi_H = 0$ in both rounds. `H` is pivotal under unanimity but is not a formal agenda setter.

What happens after terminal Round-2 rejection? The game ends. `H` receives an external disagreement payoff $d_H(\theta)$, and weak states receive their normalized disagreement payoff zero.

Is y a fixed transfer from the institutional pie? No. y is a relative institutional concession. It is always feasible as a package, but it reduces the residual payoff available to the weak coalition one-for-one.

What is being checked here? This pass resets the primitives for a fixed-pie relative-package model. All previous Round-2 formulas are treated as superseded until they are rederived from these primitives.

4 Primitives

There are $N \geq 3$ states: one hegemon H and $m = N - 1$ weak states. Nature draws

$$\theta \in \{0, 1\}.$$

The hegemon observes θ. Weak states share prior belief

$$\mu = \Pr(\theta = 1).$$

The institutional surplus available to the weak coalition is fixed and known. Normalize it to

$$B = 1.$$

The private state θ affects `H`'s participation threshold, not the size of the weak coalition's pie.

The discount factor is $\beta \in (0, 1]$.

Agenda power is represented by

$$\pi_H \in [0, 1].$$

The baseline sets

$$\pi_H = 0,$$

so only weak states are recognized as proposers. This isolates informational power through pivotality from proposal power.

5 Relative Package

A proposal is a package

$$x \in X.$$

For the first reimplementation, use a scalar concession index

$$y \in [0, \bar{y}],$$

where a higher y is more favorable to H and less favorable to the weak coalition. The package is feasible if $y \in [0, \bar{y}]$. Feasibility does not depend on θ.

The fixed-pie baseline restricts

$$0 \leq \bar{y} \leq 1.$$

Substantively, y may represent a larger production share for `H`, a weaker cut obligation, more flexibility, better enforcement against other members, or more favorable treatment under changing market conditions.

5.1 H Payoff

The minimal payoff representation is

$$U_H(y, \theta) = y + b_H(\theta). \tag{H-payoff}$$

The term $b_H(\theta)$ is the direct benefit `H` obtains from the agreement itself, apart from the concession y. It may capture cartel coordination, price stability, predictability, reputation, or the value of constraining other members.

Let

$$C_H(\theta, \mu')$$

be `H`'s continuation value after rejection, given the posterior μ'. `H` accepts package y in state θ iff

$$y + b_H(\theta) \geq \beta C_H(\theta, \mu'). \tag{H-PC}$$

Define the type-specific acceptance threshold

$$y^*_\theta(\mu') = \beta C_H(\theta, \mu') - b_H(\theta). \tag{threshold}$$

The screening condition is

$$y^*_1(\mu') > y^*_0(\mu'). \tag{screening}$$

Equivalently, in a one-shot participation version with outside option $o_H(\theta)$,

$$y^*_\theta = o_H(\theta) - b_H(\theta),$$

so screening requires

$$o_H(1) - b_H(1) > o_H(0) - b_H(0).$$

The high type requires a larger concession because its outside option net of the direct benefit of agreement is higher.

5.2 Weak-State Payoffs

We now specify how y reduces the payoff available to weak states. The normalization is one-for-one in reduced form: increasing the concession to `H` by one unit reduces the surplus available to the weak coalition by one unit.

The maintained reduced-form payoff primitive is:

$$\text{weak coalition surplus after accepted package } y = 1 - y. \tag{W-cost}$$

Thus y is costly one-for-one to the weak coalition. This is not a state-contingent feasibility constraint. The package $y \in [0, \bar{y}]$ is available in every state; if a package leaves a low residual for weak states, that is a payoff consequence, not physical infeasibility.

When weak state i is recognized and needs weak voters to approve, each non-proposing weak voter $j \neq i$ receives its continuation value c_W. The recognized weak proposer keeps the residual:

$$u_i(y, \theta; c_W) = 1 - y - (m-1)c_W, \qquad u_j(y, \theta; c_W) = c_W. \tag{W-residual}$$

In terminal Round 2, weak states' disagreement payoff is normalized to zero, so $c_W = 0$. Hence the recognized weak proposer's terminal payoff from an accepted package is $1 - y$. A representative weak state's ex ante Round-2 value is the recognized-proposer value divided by m, because each weak state is recognized with probability $1/m$ when $\pi_H = 0$.

Status: adopted primitive for the current relative-package proof pass.

6 Baseline Timing: $\pi_H = 0$

The first proof pass keeps $\pi_H = 0$ in both rounds.

1. Nature draws θ; `H` observes θ, weak states observe only μ.
2. A weak state is recognized as proposer.
3. The proposer offers a relative institutional package y.
4. Under unanimity, `H` must accept the package for agreement to form.
5. Weak voters receive continuation-value offers required for coalition approval.
6. If agreement fails, the game proceeds to the next bargaining round with beliefs updated by Bayes' rule on path.

Voting follows the standard Baron–Ferejohn convention at pivotal acceptance constraints: a player whose vote would make agreement pass and who is indifferent between accepting the agreement and receiving its continuation or outside option accepts the agreement. Thus `H` accepts at its participation threshold when it is pivotal, and weak voters accept offers equal to their continuation values when their vote is pivotal. Non-pivotal tie-breaking is specified explicitly in each voting assessment.

If a weak proposer is indifferent between two payoff-maximizing proposals, the baseline assumes that it selects the proposal that minimizes `H`'s expected payoff under the current belief. This tie-breaking rule is conservative with respect to the paper's main claim that private information allows the hegemon to extract informational rents. A later appendix should compare the baseline with the opposite efficient-agreement tie-breaking rule.

No branch in this baseline relies on a package being feasible in one state and infeasible in another.

7 Objects To Derive

7.1 R2 Unanimity

Status: fixed-pie derivation.

This subsection derives terminal Round 2 from the fixed-pie primitives above. Let

$$d_\theta = d_H(\theta), \qquad b_\theta = b_H(\theta).$$

In terminal Round 2, rejection ends the game. `H` accepts package y in state θ iff

$$y + b_\theta \geq d_\theta.$$

Define the terminal participation threshold

$$\tau_\theta = d_\theta - b_\theta.$$

The fixed-pie screening domain is

$$0 \leq \tau_0 < \tau_1 \leq \bar{y} \leq 1.$$

The proposer never benefits from choosing a package strictly inside an acceptance region, because weak surplus is $1 - y$. Therefore the only accepted packages that can be optimal are:

1. low-only: $y = \tau_0$, accepted by low `H` and rejected by high `H`;
2. pooling: $y = \tau_1$, accepted by both types.

The recognized weak proposer's payoff from low-only is

$$L_2(\mu) = (1 - \mu)(1 - \tau_0).$$

The recognized weak proposer's payoff from pooling is

$$P_2(\mu) = 1 - \tau_1.$$

The weak proposer chooses low-only iff

$$(1 - \mu)(1 - \tau_0) \geq 1 - \tau_1.$$

Solving this inequality gives the derived cutoff

$$\mu_2^* = \frac{\tau_1 - \tau_0}{1 - \tau_0}.$$

This is not a primitive. It is the belief at which the weak proposer is indifferent between the two accepted packages. Because $0 \leq \tau_0 < \tau_1 \leq 1$, $\mu_2^* \in (0, 1]$, with an interior cutoff when $\tau_1 < 1$.

Under the conservative proposer tie-breaking rule, the equilibrium Round-2 proposal is

$$y_2^*(\mu) = \begin{cases} \tau_0, & \mu \leq \mu_2^*, \\ \tau_1, & \mu > \mu_2^*. \end{cases}$$

Thus the recognized weak proposer's Round-2 value is

$$P^U_{2,prop}(\mu) = \begin{cases} (1-\mu)(1-\tau_0), & \mu \leq \mu_2^*, \\ 1-\tau_1, & \mu > \mu_2^*. \end{cases}$$

The representative weak state's Round-2 continuation value is

$$W_2^U(\mu) = \frac{P^U_{2,prop}(\mu)}{m}.$$

The high type's Round-2 continuation value is

$$C^U_{H,2}(1, \mu) = d_1$$

for all μ: it either rejects the low-only offer and receives d_1, or accepts the pooling offer at its threshold and receives d_1. The low type's Round-2 continuation value is

$$C^U_{H,2}(0, \mu) = \begin{cases} d_0, & \mu \leq \mu_2^*, \\ d_0 + \tau_1 - \tau_0, & \mu > \mu_2^*. \end{cases}$$

Status: proved for terminal Round 2 under the fixed-pie primitives, the normalized threshold domain, the Baron–Ferejohn weak-acceptance convention, and the conservative proposer tie-breaking rule. The derivation is checked in `scripts/verify_relative_package_R2_piH0.R`.

7.2 Voting Protocol for R1 Comparisons

Use the same simultaneous public voting protocol under unanimity and majority. After a proposal is made, the proposer is counted as voting yes. All other states vote simultaneously. Individual votes are publicly revealed after the ballot closes. Equivalently, voting is a secret ballot with each player's vote opened and recorded at the end.

A proposal passes if the number of yes votes meets the institutional quota:

$$Q_U = N$$

under unanimity and

$$Q_M = q = \lfloor N/2 \rfloor + 1$$

under majority. Players accept in indifference.

The public history after the ballot includes the proposal, the full vote vector, and the outcome. If the game continues, beliefs are updated from that public history.

The informational distinction comes from pivotality, not from whether a ballot is recorded. Under unanimity, `H` is pivotal because every state must vote yes. Under majority, a weak proposer may be able to assemble q weak-state yes votes without `H`; `H` may still cast a public no vote, but that vote is not pivotal on the no-screening majority path.

Therefore the majority comparison is not built by suppressing `H`'s vote. It is built by showing when a weak proposer has no reason to buy `H`'s acceptance. This is not automatic. Under majority, buying `H`'s yes vote can replace one weak voter's yes vote. The no-screening majority benchmark therefore requires a condition under which `H` is not a cheaper coalition partner than a weak voter.

Status: protocol convention adopted for the next R1 and majority derivations. The detailed assessment is recorded in `quality_reports/2026-05-11_common_voting_protocol_unanimity_majority.md`.

7.3 R1 Unanimity

Status: proved for the candidate pure-strategy equilibria below under the fixed-pie relative-package primitives, simultaneous public voting, and the stated off-path beliefs. The algebra and candidate comparisons are checked in `scripts/verify_relative_package_R1_piH0.R`.

7.3.1 R1 primitives

Round 1 uses the same players and type space as Round 2. There are $m = N - 1$ weak states and one privately informed hegemon `H`. Nature draws $\theta \in \{0, 1\}$, `H` observes θ, and weak states share the prior $\mu = \Pr(\theta = 1)$.

The baseline recognition protocol sets $\pi_H = 0$. A weak state i is recognized as proposer. The identity of the proposer is public. The proposer is counted as voting yes.

A Round-1 proposal is

$$s = (y, z_{-i}),$$

where $y \in [0, \bar{y}]$ is the relative institutional concession to `H`, and $z_j \geq 0$ is the payoff promised to each non-proposing weak voter $j \neq i$. The proposal is feasible iff

$$y + \sum_{j \neq i} z_j \leq 1. \tag{R1-feasibility}$$

This is a fixed weak-surplus constraint, not a state-contingent pie constraint. The package y is available in every state.

After the proposal, `H` and all non-proposing weak states vote simultaneously. The proposal passes under unanimity iff every state votes yes. If it passes,

$$u_H = y + b_\theta, \qquad u_j = z_j, \qquad u_i = 1 - y - \sum_{j \neq i} z_j.$$

If it fails, the game proceeds to terminal Round 2. The public history is the proposal and the full vote vector. Let ν denote the posterior probability of $\theta = 1$ after that public history. Weak states receive $\beta W_2^U(\nu)$, and type θ of H receives $\beta C_{H,2}^U(\theta, \nu)$.

From the R2 derivation, define the recognized weak proposer's terminal value

$$p_2(\nu) = \max\{(1-\nu)(1-\tau_0), 1-\tau_1\},$$

with the conservative low-only tie-breaking rule at equality. Then

$$W_2^U(\nu) = \frac{p_2(\nu)}{m}, \qquad c(\nu) = \beta W_2^U(\nu) \tag{weak-cont}$$

is the Round-1 public continuation value of a weak state after a failed proposal with posterior ν. Let

$$D_0(\nu) = C_{H,2}^U(0,\nu) = \begin{cases} d_0, & \nu \le \mu_2^*, \\ d_0 + \tau_1 - \tau_0, & \nu > \mu_2^*, \end{cases} \qquad D_1(\nu) = C_{H,2}^U(1,\nu) = d_1. \tag{H-R2-values}$$

Thus H's Round-1 rejection payoff after posterior ν is $\beta D_\theta(\nu)$. Define the high type's Round-1 threshold and the low type's high-posterior threshold:

$$a_1 = \beta d_1 - b_1, \qquad a_0^1 = \beta D_0(1) - b_0. \tag{R1-thresholds}$$

The weak R1 threshold-order domain for the current screening characterization is

$$0 \le a_0^1 \le a_1 \le \bar{y} \le 1. \tag{R1-threshold-order-domain}$$

Strict low-only separation requires $a_0^1 < a_1$. If $a_0^1 = a_1$, low-only separation is blocked but pooling at a_1 remains admissible. If $a_0^1 > a_1$, the high-threshold pooling proposal below is not admissible under the stated posterior after an off-path H no vote; that reversed-order case is outside the current screening characterization and remains blocked for this proof pass.

7.3.2 Exhaustion of pure threshold candidates

For any fixed posterior ν, H accepts a package y iff

$$y + b_\theta \ge \beta D_\theta(\nu).$$

Under the R1 separation domain, the high type's threshold is above the low type's relevant threshold. Therefore a pure threshold proposal can induce only three acceptance sets for H:

1. both types accept;
2. only the low type accepts;
3. neither type's acceptance is needed because the proposal is deliberately rejected and the game continues.

A high-only accepted proposal is impossible in this scalar package order: if a package is high enough for the high type, it is also high enough for the low type under the stated domain. Within each accepted region, any package strictly above the relevant threshold is dominated for the weak proposer by lowering y. Any weak-voter offer strictly above the relevant continuation value is also dominated. Hence the only payoff-relevant pure threshold candidates with accepted agreements are pooling and low-only. Deliberate failure gives the continuation candidate. The informative-failure lemma below rules out a profitable fourth candidate in which the proposer uses a rejected ballot only to extract information before Round 2.

7.3.3 Candidate P: pooling at the high threshold

The pooling proposal is

$$y_P = a_1, \qquad z_j = c(\mu) \quad \text{for all } j \neq i. \tag{P-proposal}$$

It is physically feasible at belief μ iff

$$a_1 + (m-1)c(\mu) \leq 1. \tag{P-feas}$$

It is an admissible pooling voting assessment iff (P-feas) holds and $a_1 \geq a_0^1$.

The proposed strategy profile is: both types of `H` vote yes; every non-proposing weak state votes yes.

Posterior beliefs after public vote vectors are:

$$\nu_P(v) = \begin{cases} \mu, & H \text{ votes yes,} \\ 1, & H \text{ votes no.} \end{cases} \tag{P-beliefs}$$

The all-yes vector is on path and the proposal passes. Vectors with `H` voting yes and at least one weak no vote are off path but preserve the prior because `H`'s prescribed yes vote is pooling. Vectors with `H` voting no are off path; the belief assigns probability one to the high type.

Weak voters' constraint is

$$z_j \geq c(\mu). \tag{P-W}$$

If a weak voter votes yes, the proposal passes and it receives z_j. If it unilaterally votes no while all others follow the proposed strategy, the proposal fails, `H`'s yes vote is uninformative, the posterior is μ, and the voter receives $c(\mu)$. With $z_j = c(\mu)$, weak voters accept in indifference.

`H`'s constraints are

$$a_1 + b_1 = \beta d_1 = \beta D_1(1), \tag{P-H1}$$

so the high type accepts by the weak-acceptance convention, and

$$a_1 + b_0 \geq \beta D_0(1), \qquad \text{equivalently} \qquad a_1 \geq a_0^1, \tag{P-H0}$$

so the low type also accepts. The latter is implied by the R1 separation domain.

The recognized weak proposer's pooling payoff is

$$\Pi_P^U(\mu) = 1 - a_1 - (m-1)c(\mu). \tag{P-payoff}$$

Voting-continuation proof. Given the beliefs in (P-beliefs), each weak voter is indifferent between yes and no at $z_j = c(\mu)$ and therefore votes yes by the weak-acceptance convention. The high type of `H` is indifferent between acceptance and rejection at $y = a_1$ and votes yes by the same convention. The low type weakly prefers acceptance because $a_1 \geq a_0^1$. The proposer pays the minimum concession and minimum weak-voter offers consistent with these constraints, so no lower-cost pooling proposal exists. Beliefs on the all-yes path are Bayesian; off-path beliefs are explicitly specified. Therefore the pooling proposal is an admissible voting-continuation assessment whenever (P-feas) holds and $a_1 \geq a_0^1$. It is a full R1 PBE only when the proposer-optimality condition stated below also selects pooling.

7.3.4 Candidate L: low-only acceptance

The low-only proposal is

$$y_L = a_0^1, \qquad z_j = c(0) \quad \text{for all } j \neq i. \tag{L-proposal}$$

It is feasible iff

$$a_0^1 + (m-1)c(0) \leq 1, \tag{L-feas}$$

and it requires the strict separation condition $a_0^1 < a_1$. The proposed strategy profile is: low `H` votes yes, high `H` votes no, and every non-proposing weak state votes yes.

Posterior beliefs after public vote vectors are:

$$\nu_L(v) = \begin{cases} 0, & H \text{ votes yes,} \\ 1, & H \text{ votes no.} \end{cases} \tag{L-beliefs}$$

Thus the all-yes vector passes and reveals the low type. The vector in which `H` votes no and all weak states vote yes is on path in the high state, fails, and reveals the high type. Off-path vectors inherit the same rule: histories with `H` yes have posterior zero, and histories with `H` no have posterior one.

Weak voters' constraint is

$$z_j \geq c(0). \tag{L-W}$$

To see this, compare a weak voter's yes and no payoffs given the proposed strategy profile. If the voter votes yes, it receives z_j in the low state and continuation $c(1)$ in the high state, because high

H rejects. If the voter votes no, the proposal fails in both states; the posterior is zero after low H's yes vote and one after high H's no vote. Hence the voter votes yes iff

$$(1-\mu)z_j + \mu c(1) \geq (1-\mu)c(0) + \mu c(1),$$

which reduces to $z_j \geq c(0)$ when $\mu < 1$. At $\mu = 1$, weak voters are not payoff-pivotal for the on-path high-state rejection and accept weakly under the same offer convention.

H's constraints are

$$a_0^1 + b_0 = \beta D_0(1), \tag{L-H0}$$

so the low type accepts by the weak-acceptance convention, and

$$a_0^1 + b_1 < \beta d_1, \qquad \text{equivalently} \qquad a_0^1 < a_1, \tag{L-H1}$$

so the high type strictly rejects.

The recognized weak proposer's low-only payoff is

$$\Pi_L^U(\mu) = (1-\mu)\{1 - a_0^1 - (m-1)c(0)\} + \mu c(1). \tag{L-payoff}$$

The second term is the proposer's continuation payoff after high H rejects and the public vote vector reveals posterior one.

Voting-continuation proof. Given (L-beliefs), each weak voter is exactly compensated for the continuation it can induce in the low state and is unaffected in the high state, so voting yes is sequentially rational. Low H is indifferent between acceptance and mimicking high rejection at posterior one and votes yes by the weak-acceptance convention. High H strictly prefers rejection because $a_0^1 < a_1$. The posterior after H's on-path no vote is one by Bayes, and the posterior after H's on-path yes vote is zero by Bayes. Off-path beliefs are explicitly stated in (L-beliefs). The proposer uses the minimum concession and weak-voter offers consistent with low-only acceptance. Therefore the low-only proposal is an admissible voting-continuation assessment whenever (L-feas) and $a_0^1 < a_1$ hold. It is a full R1 PBE only when the proposer-optimality condition stated below also selects low-only.

7.3.5 Candidate R: no-information rejection/continuation

The continuation proposal designates one non-proposing weak state r as a strict rejecter and sets

$$y_R = a_1, \qquad z_j = 0 \quad \text{for all } j \neq i. \tag{R-proposal}$$

It is feasible when $a_1 \leq \bar{y} \leq 1$. It is admissible as a strict rejection assessment at beliefs with $c(\mu) > 0$. The proposed voting strategy is: both types of H vote yes, the designated weak state r votes no, and all other non-proposing weak states vote yes. The proposal fails under unanimity because one weak state votes no.

The posterior after every failed vote vector following this proposal is

$$\nu_R(v) = \mu. \tag{R-beliefs}$$

On path, H's yes vote is pooling and weak votes carry no private information, so this posterior is Bayesian. Off path, the same belief rule keeps unilateral vote deviations from changing continuation beliefs.

The designated weak state r strictly prefers no when $c(\mu) > 0$. If it votes no, the proposal fails and it receives $c(\mu)$. If it unilaterally votes yes while all others follow the proposed strategy, the proposal passes and it receives $z_r = 0$. All other non-proposing weak states are non-pivotal because r votes no; they vote yes by the non-pivotal tie-breaking rule.

H is not outcome-pivotal because r rejects. Under (R-beliefs), a unilateral H deviation from yes to no does not change the posterior, so both types of H are sequentially rational in voting yes. The choice $y_R = a_1$ also means that if the weak rejection were removed, the high type would be at its acceptance threshold and the low type would weakly accept.

The recognized weak proposer's continuation payoff is

$$\Pi_R^U(\mu) = c(\mu). \tag{R-payoff}$$

Voting-continuation proof. The on-path rejected vote vector preserves the prior by Bayes. The designated weak rejecter strictly prefers no when $c(\mu) > 0$. The other weak voters' yes votes are sequentially rational because their unilateral votes do not affect the outcome. H's yes vote is sequentially rational because H is not outcome-pivotal and the stated off-path beliefs make vote deviations payoff-irrelevant. The proposal therefore implements no-information continuation as an admissible voting-continuation assessment whenever $a_1 \le \bar{y}$ and $c(\mu) > 0$. It is a full R1 PBE only when the proposer-optimality condition stated below also selects rejection. This candidate is not a uniqueness claim about the voting subgame after a deliberately unacceptable proposal.

7.3.6 Lemma: informative rejected ballots do not add a profitable candidate

An informative rejected ballot would try to make the proposal fail in both states while inducing different H votes, so that the public vote vector changes the posterior before Round 2. Such a ballot cannot strictly improve on the no-information continuation candidate under the maintained domain.

To see this, suppose a rejected ballot separates H's types. Since weak voters do not observe θ, only H's vote can carry type information. In any separating rejected ballot, one H action must induce posterior zero and the other must induce posterior one. The low type's continuation payoff from revealing the low posterior is

$$\beta D_0(0) = \beta d_0.$$

The low type's payoff from mimicking the high-posterior vote is

$$\beta D_0(1) \ge \beta D_0(0),$$

with strict inequality whenever $\tau_1 < 1$. Therefore the low type does not strictly prefer to reveal itself in a rejected ballot; when the inequality is strict, separation is not incentive-compatible. When $\tau_1 = 1$, the inequality is an equality and the Round-2 weak-proposer value is linear:

$$p_2(\nu) = (1-\nu)(1-\tau_0) \quad \text{for all } \nu \in [0,1].$$

Thus even if an informative rejected ballot is sustained by indifference at the boundary, it gives the weak proposer

$$(1-\mu)c(0) + \mu c(1) = c(\mu),$$

exactly the no-information continuation payoff. Pooling rejected ballots also preserve the prior and give $c(\mu)$. Hence rejected ballots add no candidate payoff above $\Pi_R^U(\mu)$.

Status: proved under the fixed-pie R2 continuation values and simultaneous public vote records.

7.3.7 Selected pure-threshold PBE selection

The preceding subsections establish the voting-continuation assessment for each candidate. A full R1 PBE also requires proposer optimality.

For each μ, define the admissible candidate set

$$\mathcal{K}(\mu) \subseteq \{P, L, R\}$$

as follows: $P \in \mathcal{K}(\mu)$ iff (P-feas) and $a_1 \geq a_0^1$ hold; $L \in \mathcal{K}(\mu)$ iff (L-feas) and $a_0^1 < a_1$ hold; and $R \in \mathcal{K}(\mu)$ iff $a_1 \leq \bar{y}$ and $c(\mu) > 0$. Inadmissible candidates receive payoff $-\infty$. Let

$$\bar{\Pi}^U(\mu) = \max_{k \in \mathcal{K}(\mu)} \Pi_k^U(\mu). \tag{R1-max}$$

The proposer-optimal candidates are

$$\mathcal{A}(\mu) = \{k \in \mathcal{K}(\mu) : \Pi_k^U(\mu) = \bar{\Pi}^U(\mu)\}. \tag{R1-argmax}$$

If $\mathcal{A}(\mu)$ contains more than one candidate, the maintained tie-breaking rule selects the candidate that minimizes `H`'s expected payoff under the prior μ. These expected `H` payoffs are

$$H_P(\mu) = (1-\mu)(a_1 + b_0) + \mu(a_1 + b_1),$$

$$H_L(\mu) = (1-\mu)(a_0^1 + b_0) + \mu\beta d_1,$$

and

$$H_R(\mu) = (1-\mu)\beta D_0(\mu) + \mu\beta d_1.$$

Thus the baseline selected candidate is

$$k^*(\mu) \in \arg\min_{k \in \mathcal{A}(\mu)} H_k(\mu). \tag{R1-tie-break}$$

For any selected $k^*(\mu)$, the strategy profile is the corresponding candidate proposal and voting strategy above, with the stated posterior map after the public vote vector. The candidate proof gives responder sequential rationality and Bayesian consistency. The argmax condition gives proposer optimality. The informative-failure lemma rules out rejected-ballot deviations that would use public voting only to generate a better posterior. Therefore the selected assessment is a full R1 PBE within the pure threshold class generated by the fixed-pie scalar package order.

7.3.8 Selected pure-threshold PBE value under R1 unanimity

For each prior μ, define invalid candidate values as $-\infty$. The recognized weak proposer's selected pure-threshold R1 unanimity PBE value is

$$P^U_{1,prop}(\mu) = \bar{\Pi}^U(\mu), \tag{R1-U-value}$$

where admissibility is defined in $\mathcal{K}(\mu)$ above. The representative weak state's ex ante Round-1 unanimity value is

$$W^U_1(\mu) = \frac{P^U_{1,prop}(\mu)}{m},$$

because each weak state is recognized with probability $1/m$ under $\pi_H = 0$.

The important correction relative to any informal low-only intuition is that the low type's R1 acceptance threshold is not $\beta d_0 - b_0$. In a separating low-only proposal, a low type that rejects mimics the high type in the public vote record and receives the high-posterior R2 continuation value. The correct low-only concession is therefore $a^1_0 = \beta D_0(1) - b_0$. If this reaches a_1, low-only separation is blocked.

Status ledger for R1 unanimity:

Object	Status	Notes
R1 primitives	Proved	Players, types, proposer identity, proposal space, voting rule, payoffs, and continuation values are explicit.
Candidate exhaustion	Proved within pure threshold class	Accepted proposals reduce to pooling or low-only; informative rejected ballots do not improve on no-information continuation.

Object	Status	Notes
Pooling candidate	Proved and checked	Admissible voting-continuation assessment under (P-feas) and $a_1 \geq a_0^1$; full PBE when selected by the argmax/tie-break rule.
Low-only candidate	Proved and checked	Admissible voting-continuation assessment under (L-feas) and strict $a_0^1 < a_1$; blocked when $a_0^1 \geq a_1$; full PBE when selected.
Rejection/continuation candidate	Proved and checked	Admissible no-information continuation assessment when $a_1 \leq \bar{y}$ and $c(\mu) > 0$; full PBE when selected.
Selected pure-threshold PBE selection	Proved and checked	Selected candidates maximize the proposer payoff among admissible candidates; ties minimize `H`'s expected payoff.
Global uniqueness of all R1 equilibria	Pending	The result characterizes the selected pure-threshold PBE value, not every weak voting equilibrium after arbitrary off-path proposals. Mixed or semi-pooling rejected ballots remain outside the current claim.
Numerical and algebraic verification	Checked	See `scripts/verify_relative_package_R1_piH`

7.4 Majority

Let

$$q = \lfloor N/2 \rfloor + 1, \qquad k = q - 1.$$

The recognized weak proposer is counted as voting yes, so under majority it needs k additional yes votes.

7.4.1 R2 majority

In terminal Round 2, weak voters' outside payoff is zero. The proposer can set $y = 0$, offer zero to k weak voters, and pass the proposal without `H`. Any $y > 0$ is costly to the weak coalition and is not needed for passage. Thus the recognized weak proposer's terminal majority value is

$$P^M_{2,prop} = 1.$$

The representative weak state's terminal majority value is

$$W_2^M = \frac{1}{m}.$$

If excluded under majority, H receives its external payoff d_θ, outside the weak coalition's institutional surplus:

$$C_{H,2}^M(\theta) = d_\theta.$$

7.4.2 R1 majority

The public continuation value of a weak voter after R1 rejection is

$$c_M = \beta W_2^M = \frac{\beta}{m}.$$

A no-H majority proposal sets $y = 0$, offers c_M to k weak voters, and offers zero to all other states. It is accepted by the selected weak voters by the Baron–Ferejohn weak-acceptance convention. The recognized weak proposer's payoff is

$$P_{1,noH}^M = 1 - kc_M = 1 - \frac{(q-1)\beta}{m}.$$

Now check whether the proposer can do better by buying H's vote and one fewer weak vote. Any majority package that includes H and still buys k weak votes is weakly dominated by dropping the concession to H, since H's vote is not needed for passage. Any H-including package that sets y strictly inside an acceptance region is dominated by lowering y to the relevant threshold. Thus the only potentially profitable H-including alternatives are pooling at the high threshold and low-only at the low threshold, each with $k - 1$ weak voters.

If H is made pivotal in an R1 majority coalition, its type-specific threshold is

$$a_\theta^M = \beta C_{H,2}^M(\theta) - b_H(\theta) = \beta d_\theta - b_H(\theta).$$

Assume the R1 majority threshold domain

$$0 \leq a_0^M < a_1^M \leq \bar{y}.$$

A pooling H-including majority proposal costs $a_1^M + (k-1)c_M$. It is not better than the no-H proposal iff

$$a_1^M \geq c_M.$$

A low-only H-including majority proposal costs $a_0^M + (k-1)c_M$ when $\theta = 0$, but fails and yields continuation value c_M to the proposer when $\theta = 1$. Its expected payoff is

$$P^M_{1,LH}(\mu) = (1-\mu)\{1 - a^M_0 - (k-1)c_M\} + \mu c_M.$$

Relative to the no-`H` proposal,

$$P^M_{1,LH}(\mu) - P^M_{1,noH} = c_M - a^M_0 + \mu(a^M_0 + kc_M - 1). \qquad \text{(majority-low-H-gap)}$$

Since $q = k+1 \leq m$ for $N \geq 3$, and $\beta \leq 1$,

$$(k+1)c_M = \frac{q\beta}{m} \leq 1. \qquad \text{(quota-bound)}$$

Therefore, if

$$a^M_0 \geq c_M. \qquad \text{(no-cheap-H)}$$

then $P^M_{1,LH}(0) - P^M_{1,noH} \leq 0$ and $P^M_{1,LH}(1) - P^M_{1,noH} = (k+1)c_M - 1 \leq 0$. The gap is linear in μ, so the low-only `H`-including proposal is weakly dominated by the no-`H` proposal for every $\mu \in [0,1]$.

Because $a^M_1 > a^M_0$, condition (no-cheap-H) also implies $a^M_1 \geq c_M$, so pooling purchases of `H`'s vote are not profitable. Under this condition, the R1 majority equilibrium path has $y = 0$, a weak-state winning coalition, and no screening of `H`.

The condition is also necessary for uniform no-screening. If $a^M_0 < c_M$, then at $\mu = 0$

$$P^M_{1,LH}(0) - P^M_{1,noH} = c_M - a^M_0 > 0.$$

By continuity, a low-only `H`-including proposal strictly beats the no-`H` proposal for sufficiently low beliefs that `H` is high. In that region, majority itself has a screening motive: the weak proposer buys low `H` as a cheap substitute for one weak voter and lets the proposal fail when `H` is high. The cutoff for this majority-screening region is

$$\mu^H_M = \frac{c_M - a^M_0}{1 - a^M_0 - kc_M}$$

whenever $a^M_0 < c_M$. The quota bound above implies $\mu^H_M \in (0,1]$.

Thus condition (no-cheap-H) is not a verbal regularity condition. It is necessary and sufficient for the no-`H` majority proposal to weakly dominate `H`-including majority proposals for all beliefs.

The recognized weak proposer's R1 majority value is therefore

$$P^M_{1,prop} = 1 - \frac{(q-1)\beta}{m},$$

and the symmetric ex ante representative weak-state value is

$$W^M_1 = \frac{1}{m}.$$

Status: derived under the simultaneous public voting protocol and the no-cheap-`H` condition. The derivation is checked in `scripts/verify_relative_package_majority_piH0.R`.

7.5 Entry and Institutional Choice

After R2, R1, and majority are rederived, rebuild:

1. weak-state entry and nesting;
2. conditional institutional comparison for `H`;
3. calibrated OPEC classification.

8 Reproducibility Plan

Create new scripts rather than modifying the archived C-B-R scripts in place. Current and planned scripts:

1. `scripts/verify_relative_package_R2_piH0.R` (created and run for fixed-pie R2)
2. `scripts/verify_relative_package_R1_piH0.R`
3. `scripts/verify_relative_package_majority_piH0.R`
4. `scripts/verify_relative_package_entry_nesting_piH0.R`
5. `scripts/verify_relative_package_classification_piH0.R`

Each result should be independently verified before being promoted in the proof ledger.

9 Proof Ledger

Object	Status	Notes
Archived feasibility-branch derivation	Archived	Preserved in tag `redesign-feasibility-branch-2026-05-11`; do not use as current architecture.
Relative-package architecture	Adopted	Packages are feasible in every state; screening comes from type-dependent participation thresholds.
Minimal H payoff primitive	Adopted	$U_H(y,\theta) = y + b_H(\theta)$.
Simultaneous public voting protocol	Adopted for next derivations	The proposer is counted as voting yes; all other states vote simultaneously; individual votes are publicly revealed after the ballot; unanimity uses quota N, majority uses quota $q = \lfloor N/2 \rfloor + 1$.
Screening condition	Adopted as target primitive	Need $y_1^*(\mu') > y_0^*(\mu')$, equivalently $\beta C_H(1,\mu') - b_H(1) > \beta C_H(0,\mu') - b_H(0)$.

Object	Status	Notes
Weak-state cost of concession	Adopted for fixed-pie reset	y is costly one-for-one to the weak coalition; accepted weak surplus is $1-y$; weak proposer keeps residual after weak voters receive continuation values.
R2 unanimity under relative packages	Proved under fixed-pie baseline; checked numerically	Weak proposer chooses low-only for $\mu \le \mu_2^*$ and pooling for $\mu > \mu_2^*$, where $\mu_2^* = (\tau_1 - \tau_0)/(1-\tau_0)$.
R1 unanimity under relative packages	Proved selected pure-threshold PBE value; checked numerically	Pooling, low-only, and rejection/continuation are derived from simultaneous public voting; full PBE selection requires proposer argmax and the H-payoff tie-break. Low-only is blocked when $a_0^1 \ge a_1$. Checked in `scripts/verify_relative_package_R1_piH(`
Majority under relative packages	Derived and checked numerically	No screening under $\pi_H = 0$ requires the no-cheap-`H` condition $a_0^M \ge \beta/m$; otherwise `H` may be a cheap majority coalition partner.
Entry/nesting	Pending	Rebuild after R1 and majority.
Institutional classification	Pending	Rebuild after entry and conditional comparison.
$\pi_H > 0$ extensions	Pending	Treat agenda power separately after the baseline is verified.